

Hardware Design and Verification of Dilithium – Post Quantum Digital Signature Algorithm

M.Engg IC Design Thesis by

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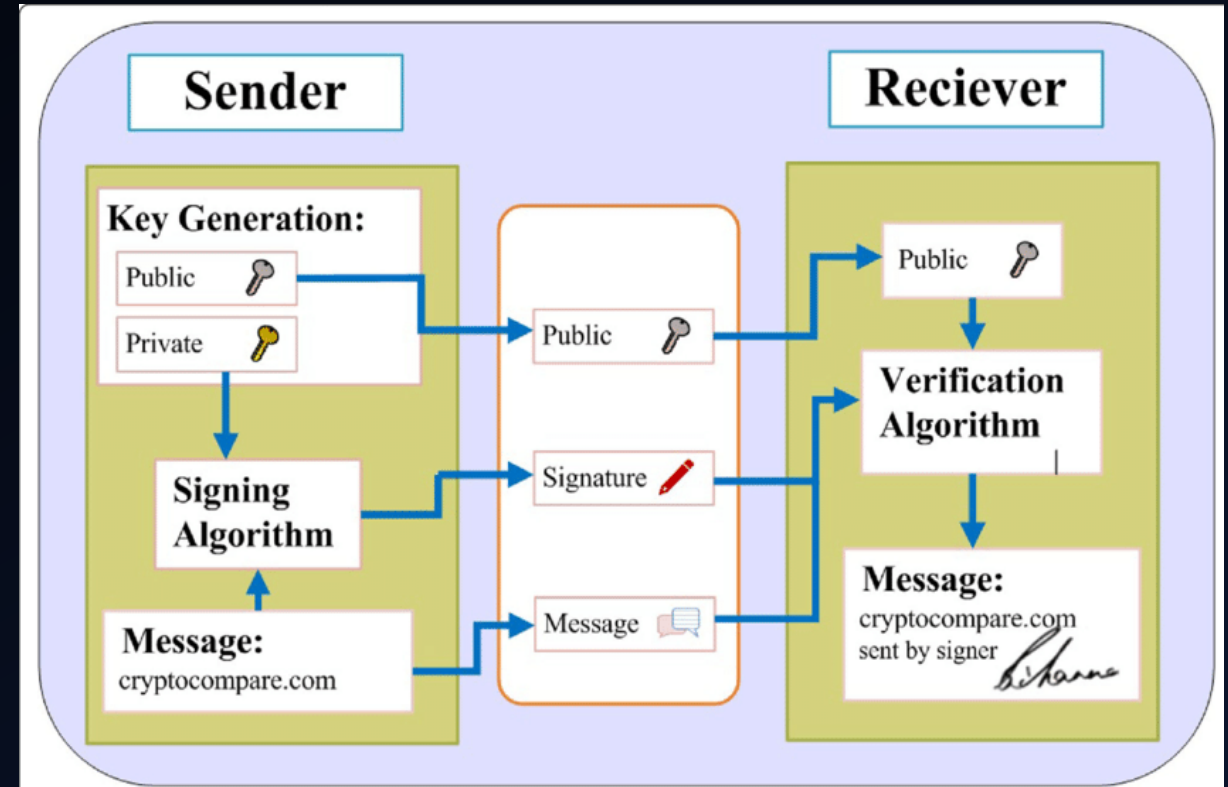
DR . SONIYA SHAFI

Background

- Data security demands are rapidly increasing.
- Quantum computers threaten current cryptographic algorithms.
- Sensitive data can be harvested today and decrypted later.
- Post-quantum cryptography is designed to provide long-term security against quantum attacks.

WHAT IS DILITHIUM?

- Dilithium is a module lattice based post-quantum digital signature algorithm.
- A digital signature ensures three things:
 - Authentication – Verifies who sent the message.
 - Integrity – Ensures the message was not changed.
 - Non-repudiation – Sender cannot deny sending the message.
- Dilithium Operations
 - Key generation
 - Message signing
 - Verification



PROBLEM STATEMENT

Software based implementations of CRYSTALS Dilithium suffer from high computational complexity, increased latency, and elevated power consumption

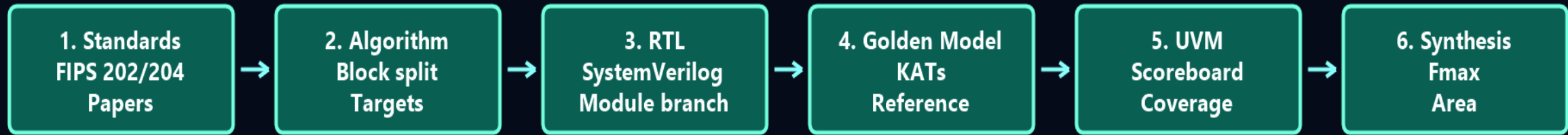
OBJECTIVE

Design and verify an efficient hardware architecture for Dilithium, implement key modules (polynomial arithmetic & NTT), and evaluate latency, area, and power.

Hardware Literature Review

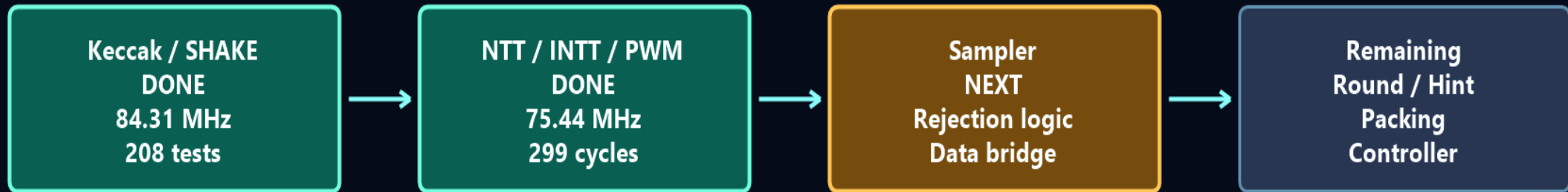
Year	Work / Paper	Focus	Key point
2021	[1] Beckwith et al. - High-Performance CRYSTALS-Dilithium	Full FPGA accelerator	High-performance baseline for complete Dilithium hardware.
2022	[2] Zhao et al. - Compact High-Performance Dilithium	Compact accelerator	Area-conscious architecture and resource/latency trade-offs.
2023	[3] Mao et al. - Configurable SW/HW Co-Design	Configurable co-design	Hardware/software split for performance and flexibility.
2024	[4] Wang et al. - RISC-V SoC FPGA Co-Design	SoC integration	Reference for connecting PQC accelerators to a processor system.
2025	[5] Norga et al. - ML-DSA-OSH Open-Source Hardware	Open-source accelerator	Reproducible modern full ML-DSA hardware reference.
2025	[6] Cui et al. - Improved MDC-NTT CRYSTALS-Dilithium	Streaming / MDC architecture	Useful Dilithium hardware comparison for high-throughput polynomial datapath.
2025	[7] Nguyen and Nguyen - Area-Time Efficient Dilithium	Area-time design	Useful area-time comparison and 2x2 NTT architecture reference.
2026	[8] Chowdhury et al. - Unified Kyber/Dilithium Fault Tolerance	Unified secure accelerator	System-level Kyber/Dilithium design with fault-tolerance direction.
2026	[9] Ni et al. - LightHD CRYSTALS-Dilithium Accelerator	Lightweight accelerator	Recent system-level design emphasizing resource efficiency.
2025	[10] Sun et al. - PALS High-Speed ML-DSA Pipeline	Pipeline integration	Most relevant next step: sampler-to-NTT reordering and full integration.

Methodology

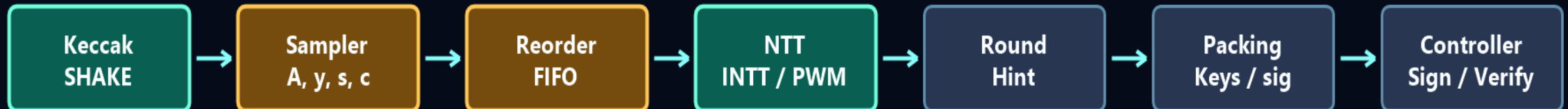


Where We Are Now

Keccak and NTT are complete. Next is sampler and dataflow integration.



Planned Full ML-DSA Dataflow



■ Done ■ Next milestone ■ Remaining work

Dilithium High-Level Architecture

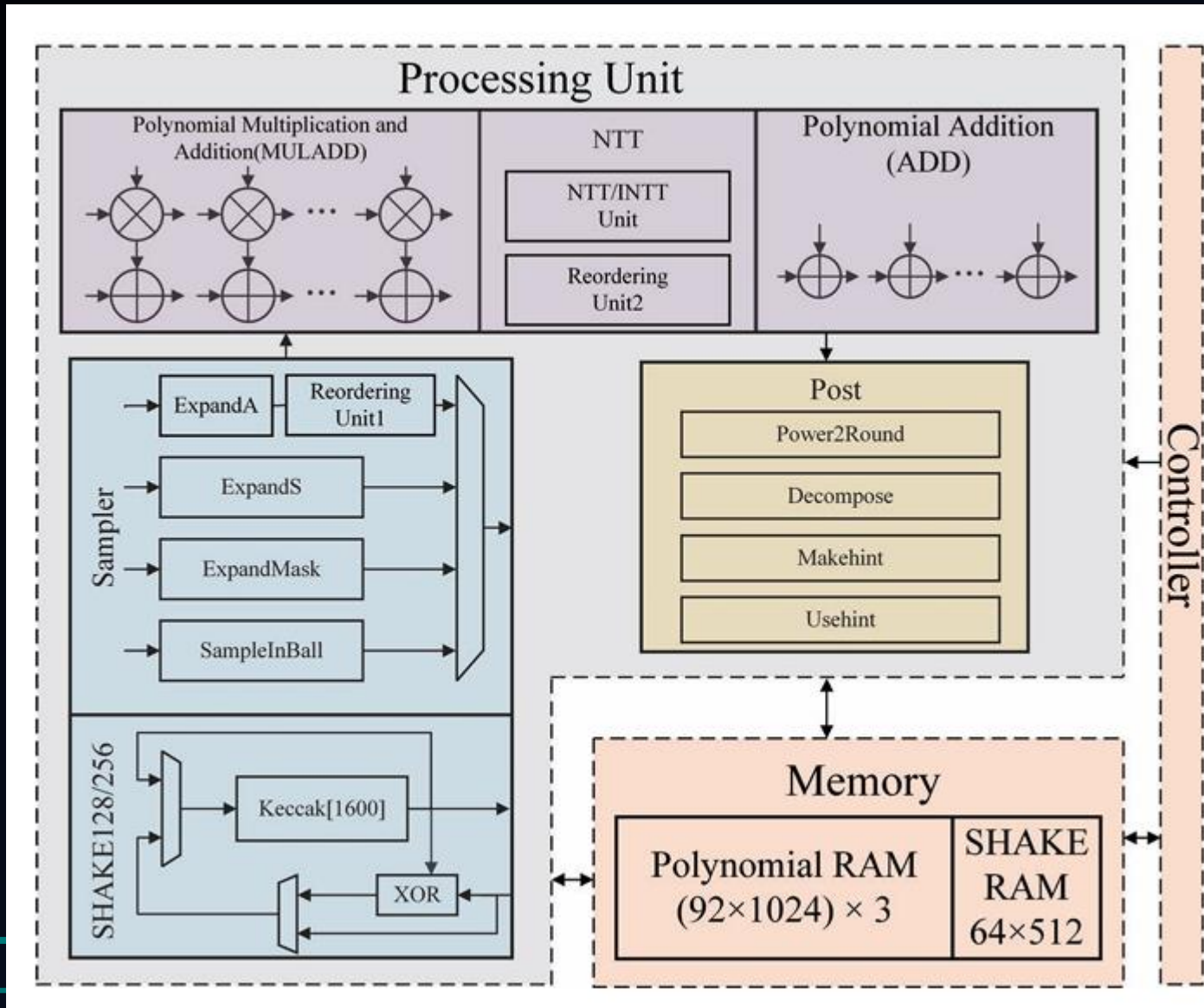


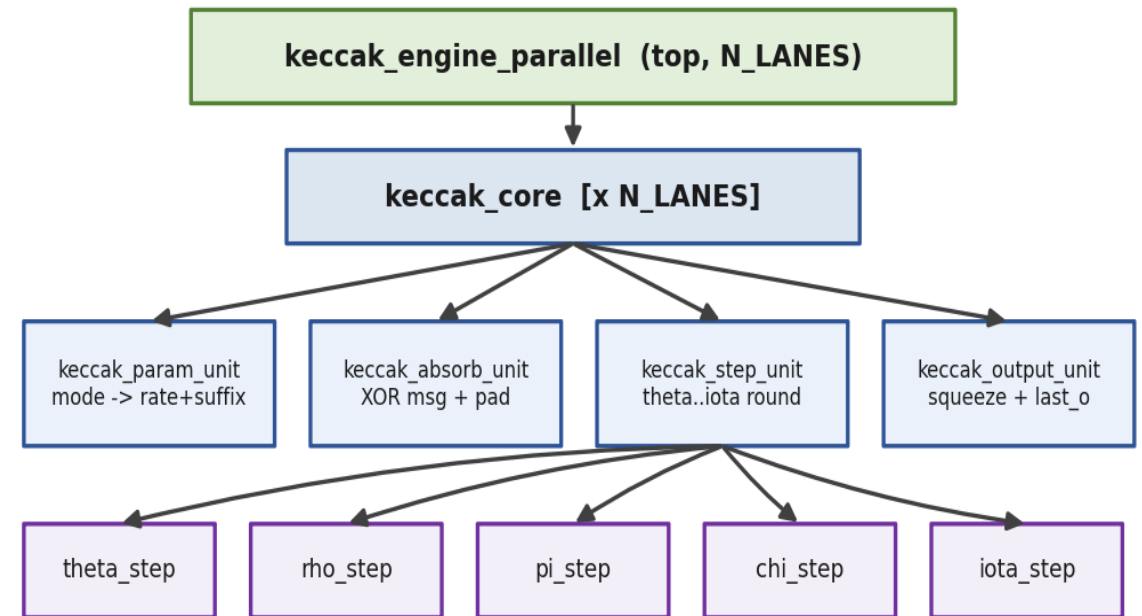
Fig. Dilithium High-Level Architecture. Courtesy- PALS 2025 [10]

Dilithium Major Components

- Keccak engine (SHAKE-128/256) (Hashing)
- Numeric Theoretic Transform (NTT) module (Poly Multiplication)
- Memory
 - Sampler
 - Rounding/decomposition logic
 - Dilithium Main Controller

Keccak Engine – Design

- Works on Sponge Construction
 - Absorb
 - Permute($\theta \rightarrow \rho \rightarrow \pi \rightarrow \chi \rightarrow \iota$)
 - Squeeze
- 24 permutation rounds (24 cyc/perm)
- SHAKE128/SHAKE256 hashing support
- 64-bit bus
- 2-stage pipelined design (48 cyc/perm)
- parameterized parallel lanes
- 4-lane speedup: 3.84x



Keccak Engine – Verification and Synthesis

- Cover group samples each completed SHAKE transaction
- Cover points:
 - SHAKE mode: SHAKE128, SHAKE256
 - message length: empty, short, rate-boundary, multi-block
 - XOF length: continuous, small, medium, large
- Cross coverage:
 - mode x message length
 - mode x XOF length
- Verified against System Verilog golden model
- Results:
 - 208/208 UVM tests passed
 - 100% functional coverage
 - Fmax: 84.31 MHz (Cyclone V)
 - ALMS: 3,747 (4 Lanes)
 - Latency: 48 cyc/perm (2-stage pipeline)

Keccak Engine – Verification and Synthesis Outputs

Covergroups						
Name	Class Type	Coverage	Goal	% of Goal	Status	Included
/tb_top_sv_unit/keccak_coverage		100.00%				
TYPE cg_keccak		100.00%	100	100.00...		
CVP cg_keccak::cp_mode		100.00%	100	100.00...		
CVP cg_keccak::cp_xof_kind		100.00%	100	100.00...		
CVP cg_keccak::cp_msg_len		100.00%	100	100.00...		
CROSS cg_keccak::cross_mode_msg		100.00%	100	100.00...		
CROSS cg_keccak::cross_shake_xof		100.00%	100	100.00...		

```
# UVM_INFO verillog_src/uvm-1.1d/src/base/uvm_object.svh(1267) @ 63955000: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_coverage.sv(49) @ 63955000: uvm_test_top.env.coverage_0 [keccak_coverage] Functional Coverage: 100.00%
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_coverage.sv(49) @ 63955000: uvm_test_top.env.coverage_1 [keccak_coverage] Functional Coverage: 100.00%
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_coverage.sv(49) @ 63955000: uvm_test_top.env.coverage_2 [keccak_coverage] Functional Coverage: 100.00%
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_coverage.sv(49) @ 63955000: uvm_test_top.env.coverage_3 [keccak_coverage] Functional Coverage: 100.00%
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_scoreboard.sv(84) @ 63955000: uvm_test_top.env.scoreboard_0 [keccak_scoreboard]
===== SCOREBOARD SUMMARY =====
# Total: 52
# Passed: 52
# Failed: 0
===== SCOREBOARD SUMMARY =====
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_scoreboard.sv(84) @ 63955000: uvm_test_top.env.scoreboard_1 [keccak_scoreboard]
===== SCOREBOARD SUMMARY =====
# Total: 52
# Passed: 52
# Failed: 0
===== SCOREBOARD SUMMARY =====
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_scoreboard.sv(84) @ 63955000: uvm_test_top.env.scoreboard_2 [keccak_scoreboard]
===== SCOREBOARD SUMMARY =====
# Total: 52
# Passed: 52
# Failed: 0
===== SCOREBOARD SUMMARY =====
# UVM_INFO ../tb_uvm/tb_uvm_keccak_v2/keccak_scoreboard.sv(84) @ 63955000: uvm_test_top.env.scoreboard_3 [keccak_scoreboard]
===== SCOREBOARD SUMMARY =====
# Total: 52
# Passed: 52
# Failed: 0
===== SCOREBOARD SUMMARY =====
# --- UVM Report Summary ---
# ** Report counts by severity
# UVM_INFO : 637
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
```

Slow 1100mV 85C Model Fmax Summary

<<Filter>>

	Fmax	Restricted Fmax	Clock Name	Note
1	84.31 MHz	84.31 MHz	clk	

Flow Summary

<<Filter>>

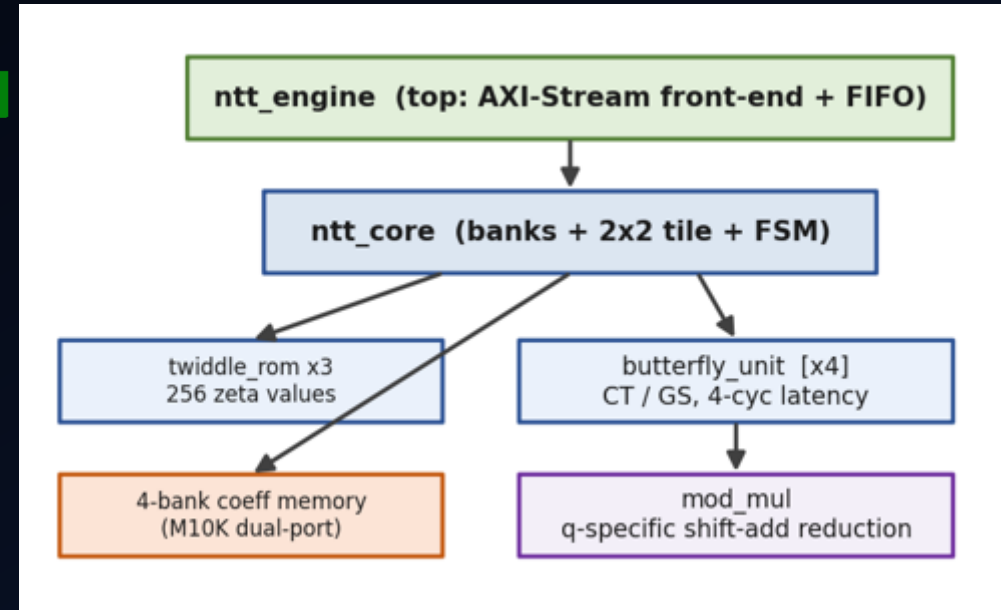
Flow Status	Successful - Tue Jun 16 03:46:59 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	keccak
Top-level Entity Name	keccak_core
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	3,738 / 56,480 (7 %)
Total registers	3270
Total pins	202 / 268 (75 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

NTT – Design

- FFT without Complex Numbers
- Implements polynomial Multiplication over

$$R_q = \mathbb{Z}_q[x] / (x^{256} + 1), \quad q = 8,380,417$$

- Supports three operations:
 - forward NTT
 - inverse NTT with scaling
 - point-wise multiplication
- Memory-based architecture with a **2x2 butterfly tile (4 Memory banks)**
- Uses **Cooley-Tukey butterflies** for forward NTT
- Uses **Gentleman-Sande butterflies** for inverse NTT
- q-specific shift-add modular reduction instead of Barrett reduction
- 299 cycles per NTT



NTT – Verification and Synthesis

- Cover group samples each completed NTT transaction
- Cover points:
 - Operation: forward NTT, inverse NTT, point-wise multiplication
 - Input pattern: zero, constant, random, boundary coefficients
 - Transform case: forward-only, inverse-only, round-trip
 - Output range: coefficients reduced modulo q
- Cross coverage:
 - operation x input pattern
 - operation x coefficient range
 - transform case x edge case
- Verified against System Verilog golden model
- Results:
 - 53/53 UVM tests passed
 - 100% functional coverage
 - Fmax: 75.44 MHz (Cyclone V)
 - ALMS: 2,313 , DSPS: 6 , M10K: 18
 - Latency 299 cyc/ntt , 3.96 us

NTT – Verification and Synthesis Outputs

Name	Coverage	Goal	% of Goal	Status	Include
/tb_top_sv_unit/ntt_coverage	100.00%				
TYPE cg_ntt	100.00%	100	100.00%		
CVP cg_ntt::cp_op	100.00%	100	100.00%		
bin ntt	15	1	100.00%		
bin intt	23	1	100.00%		
bin pwm	15	1	100.00%		
CVP cg_ntt::cp_in_kind	100.00%	100	100.00%		
bin zero	6	1	100.00%		
bin max	6	1	100.00%		
bin delta	6	1	100.00%		
bin random	35	1	100.00%		
CROSS cg_ntt::cross_op_kind	100.00%	100	100.00%		
bin <pwm,random>	9	1	100.00%		
bin <intt,random>	17	1	100.00%		
bin <ntt,random>	9	1	100.00%		
bin <pwm,delta>	2	1	100.00%		
bin <intt,delta>	2	1	100.00%		
bin <ntt,delta>	2	1	100.00%		
bin <pwm,max>	2	1	100.00%		
bin <intt,max>	2	1	100.00%		
bin <ntt,max>	2	1	100.00%		
bin <pwm,zero>	2	1	100.00%		
bin <intt,zero>	2	1	100.00%		
bin <ntt,zero>	2	1	100.00%		

```
# UVM_INFO ../tb_uvm/tb_uvm_ntt/ntt_scoreboard.sv(54) @ 580605000: uvm_test_top.env.scoreboard [ntt_scoreboard] [PASS] COV PWM delta
# UVM_INFO ../tb_uvm/tb_uvm_ntt/ntt_scoreboard.sv(54) @ 591665000: uvm_test_top.env.scoreboard [ntt_scoreboard] [PASS] COV PWM random
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 591865000: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
# UVM_INFO ../tb_uvm/tb_uvm_ntt/ntt_coverage.sv(40) @ 591865000: uvm_test_top.env.coverage [ntt_coverage] Functional Coverage: 100.00%
# UVM_INFO ../tb_uvm/tb_uvm_ntt/ntt_scoreboard.sv(68) @ 591865000: uvm_test_top.env.scoreboard [ntt_scoreboard]
# ===== SCOREBOARD SUMMARY =====
# Total: 53
# Passed: 53
# Failed: 0
# =====
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 60
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [UVMTOP] 1
# [ntt_coverage] 1
# [ntt_scoreboard] 54
# ** Note: $finish : C:/questasim64_2024.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 591865 ns Iteration: 54 Instance: /tb_top
```

Slow 1100mV 85C Model Fmax Summary

<<Filter>>

	Fmax	Restricted Fmax	Clock
1	75.44 MHz	75.44 MHz	clk

Flow Summary

<<Filter>>

Flow Status	Successful - Tue Jun 16 04:43:35 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	ntt
Top-level Entity Name	ntt_engine
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	2,313 / 56,480 (4 %)
Total registers	951
Total pins	59 / 268 (22 %)
Total virtual pins	0
Total block memory bits	30,333 / 7,024,640 (< 1 %)
Total DSP Blocks	6 / 156 (4 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Comparison with published designs

NTT

Paper (year) [ref]	Device	Fmax	Area / logic	Architecture	Latency
This work (2026)	Cyclone V	75.44 MHz	2,313 ALM; 6 DSP; 18 M10K	2x2 BF; 4-bank; shift-add mod	299 cyc; 3.96 us
Compact Dilithium (2022) [2]	Artix-7	96.9 MHz	29,998 LUT full	4 BFU; folding	533 cyc; 5.50 us
ML-DSA-OSH (2025) [5]	ZCU102 US+	230 MHz	54,942 LUT full	2x2 BF; CT/GS; Barrett	N/R
Improved MDC-NTT (2025) [6]	Artix-7	297 MHz	4,187 LUT; 18 DSP	radix-2 MDC; 8 BF	286 cyc; 0.96 us
Area-Time Efficient (2025) [7]	Versal	328 MHz	55.6k LUT full	2x2 BF; addr resolver	~256 cyc; 0.78 us
LightHD (2026) [9]	Artix-7	212.8 MHz	2,601 LUT; 8 DSP	iterative NTT; pre-NTT LUT	235-264 cyc
PALS (2025) [10]	A7 / US+	103 / 251 MHz	33,799 LUT full	sampler-NTT pipeline	L5 sign: 312/128 us
EMINEM (2025) [12]	Kintex US+	324 MHz	1,552 LUT; 6 DSP	mixed-radix / radix-4	271 cyc; 0.84 us
PQShield NTT (2024) [13]	Zynq US+	322 MHz	3,821 LUT; 20 DSP	R2MDC streaming	258 cyc; 0.80 us
Conflict-Free NTT (2026) [14]	ZCU102 US+	332 / 308 MHz	1,423/3,185 LUT	PTM SPRAM; 1/2 BF	1075 / 563 cyc

KECCAK

Paper (year) [ref]	Device	Fmax	Area / logic	Architecture	Latency
This work (2026)	Cyclone V	84.31 MHz	3,747 ALM; 0 DSP/BRAM	SHAKE128/256; 2-stage round; 4 lanes	48 cyc/perm
High-Performance Dilithium (2021) [1]	A7 / K7 / VUS+	256 MHz	5,483 LUT x3	3 Keccak cores	N/R
Compact Dilithium (2022) [2]	Artix-7	96.9 MHz	15,784 LUT	2 rounds/cyc	12 cyc/perm
ML-DSA-OSH (2025) [5]	ZCU102 US+	230 MHz	18,678 LUT	3 SHAKE cores	N/R
LightHD (2026) [9]	Artix-7	212.8 MHz	6,019 LUT	dual Keccak; 26-cyc offset	25/24/25 cyc
PALS (2025) [10]	A7 / US+	103 / 251 MHz	33,799 LUT full	SHAKE-sampler pipeline	L5 sign: 312/128 us
SHA-3 Artix-7 (2026) [11]	Artix-7	210 MHz	24% LUT; 5% DSP	24-stage SHA-3 pipeline	1.35 Gbps

Remaining Work

- Sampler
- Rounding/decomposition logic
- Dilithium Main Controller and Module Integration
- Complete Dilithium Verification and Synthesis
- end-to-end ML-DSA KAT verification

GANTT CHART

Work Breakdown	Months											
	TH 1						TH 2					
	1	2	3	4	5	6	7	8	9	10	11	12
Literature study and algorithm understanding												
Requirement analysis and architecture design												
Design and Verification of Hashing module												
Design and Verification of polynomial arithmetic module												
Design and Verification of Number Theoretic Transform module												
Complete hardware design Testbench development and verification												
Performance evaluation and optimization												
Documentation and thesis writing												

References

- [1] L. Beckwith, D. T. Nguyen, and K. Gaj, “High-Performance Hardware Implementation of CRYSTALS-Dilithium,” International Conference on Field-Programmable Technology, 2021.
- [2] C. Zhao, N. Zhang, H. Wang, B. Yang, W. Zhu, Z. Li, M. Zhu, S. Yin, S. Wei, and L. Liu, “A Compact and High-Performance Hardware Architecture for CRYSTALS-Dilithium,” IACR Transactions on Cryptographic Hardware and Embedded Systems, vol. 2022, no. 1, pp. 270–295, 2022.
- [3] G. Mao, D. Chen, G. Li, W. Dai, A. I. Sanka, C. K. Koc, and R. C. C. Cheung, “High-performance and Configurable SW/HW Co-design of Post-quantum Signature CRYSTALS-Dilithium,” ACM Transactions on Reconfigurable Technology and Systems, vol. 16, no. 3, Article 44, 2023.
- [4] T. Wang, C. Zhang, X. Zhang, D. Gu, and P. Cao, “Optimized Hardware-Software Co-Design for Kyber and Dilithium on RISC-V SoC FPGA,” IACR Transactions on Cryptographic Hardware and Embedded Systems, vol. 2024, no. 3, pp. 99–135, 2024.
- [5] Q. Norga, S. Kundu, and I. Verbaauwhede, “ML-DSA-OSH: An Efficient, Open-Source Hardware Implementation of ML-DSA,” full version, 2025.
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- [9] Z. Ni, A. Khalid, Z. Zhang, Y. Cui, W. Liu, and M. O’Neill, “LightHD: A Lightweight and High-Performance Hardware Accelerator of CRYSTALS-Dilithium,” IEEE Transactions on Computers, vol. 75, no. 5, pp. 2007–2019, May 2026.
- [10] P. Sun, Q. Wang, and M. Ikeda, “PALS: A High-Speed ML-DSA Hardware Accelerator with Enhanced Pipeline on FPGA,” IEEE International Conference on Integrated Circuits, Technologies and Applications, 2025.
- [11] A. A. Abdulsamad and S. R. Repas, “Design of an Energy-Efficient SHA-3 Accelerator on Artix-7 FPGA for Secure Network Applications,” *Computers*, vol. 15, no. 1, article 3, 2026.
- [12] Y. Tu and J. Xie, “EMINEM: Efficient FPGA Implementation of Mixed-Radix NTT Hardware Accelerators for NIST Post-Quantum Cryptography Falcon, Dilithium, and HAWK,” *ACM Transactions on Reconfigurable Technology and Systems*, vol. 18, no. 4, article 52, 2025.
- [13] D. E. S. Kundi, J. M. Bermudo Mera, P.-Y. Strub, and M. Hutter, “High-Performance NTT Hardware Accelerator to Support ML-KEM and ML-DSA,” *Workshop on Attacks and Solutions in Hardware Security*, pp. 100–105, 2024.
- [14] H. S. Ogawa, T. B. Paiva, M. A. Simplicio Jr., S. M. Hafiz, and B. Yildiz, “Efficient Conflict-Free NTT Hardware Architecture with Single-Port RAMs: Applications to ML-DSA,” *IACR Cryptology ePrint Archive*, Paper 2026/621, 2026.

THANK YOU ! 😊